

MODULE 09: LC CIRCUIT & MAXWELL EQUATIONS

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LECTURE 22

OUTLINE:

- Maxwell's equations

MAXWELL'S EQUATIONS

Maxwell's equations are a set of four fundamental equations that describe the behavior of electric and magnetic fields and their interactions with charges and currents. Although, Maxwell didn't invent all four laws, he made two key contributions:

- Displacement Current: He modified Ampère's Law to include the term displacement current, $i_d = \epsilon_0 \frac{dE}{dt}$, which accounts for changing electric fields (crucial for electromagnetic wave propagation).
- Unification & Symmetry: He combined the four laws into a coherent mathematical framework, showing that they describe a unified electromagnetic field.
- Equations describe all microscopic electromagnetic phenomena (scale: ~10nm to light-years).

Table: Maxwell's equations

NAME	INTEGRAL FORM	DIFFERENTIAL FORM
Gauss's Law for Electric Fields	$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$	$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$
Gauss's Law for Magnetic Fields	$\oint \vec{B} \cdot d\vec{A} = 0$	$\nabla \cdot \vec{B} = 0$
Faraday's Law of Electromagnetic Induction	$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
Ampere-Maxwell Current Law	$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} + \mu_0 \epsilon_0 \frac{d\Phi_e}{dt}$	$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$

NOTE:

- Divergence of $\vec{E}$ is written as $\nabla \cdot \vec{E}$. Physically, divergence measures how much a vector field "spreads out" or "originates" from a point. A positive divergence indicates a source (field lines coming out), while a negative divergence indicates a sink (field lines going in).
- Curl of $\vec{E}$ is written as $(\nabla \times \vec{E})$. Physically, curl measures the tendency of the field to circulate or rotate around a point. Curl vector is defined at a point, and point along the axis of rotation. Its direction given by the right-hand rule.
- Del or nabla (∇) is a three-dimensional differential mathematical operator. It is denoted by ∇ .

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Maxwell's equations are as follows in detail

Gauss's Law for Electric Fields: This equation relates the electric flux through a closed surface to the total electric charge enclosed within that surface. It is given by,

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

where $\nabla \cdot \vec{E}$ represents the divergence of the electric field $\vec{E}$, ρ is the charge density, and ϵ_0 is the vacuum permittivity.

- In integral form: $\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$
- The total electric flux through any closed surface equals to the net charge inside that surface divided by ϵ_0 .
- This relates an electric field to the charge distribution that creates it.

Gauss's Law for Magnetic Fields: This equation states that there are no magnetic monopoles and that the magnetic field lines are always closed loops. It is expressed as:

$$\nabla \cdot \vec{B} = 0$$

where $\nabla \cdot \vec{B}$ represents the divergence of the magnetic field $\vec{B}$.

- In integral form: $\oint \vec{B} \cdot d\vec{A} = 0$
- The total magnetic flux through any closed surface is zero.
- This says that number of field lines that enter a closed volume must be equal the number that leave that volume.
- This implies that the magnetic field lines cannot begin or end at any point.
- Isolated magnetic monopoles have not been observed in nature.

Faraday's Law of Electromagnetic Induction: This equation describes how a changing magnetic field induces an electric field. It is given by:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

where $\nabla \times \vec{E}$ represents the curl of the electric field $\vec{E}$, and $\frac{\partial \vec{B}}{\partial t}$ is the partial derivative of the magnetic field $\vec{B}$ with respect to time.

- In integral form: $\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$
- This describes the creation of a circulating electric field around a region in the space due to a time-varying magnetic flux.
- The law states that the electromotive force (EMF), which is the line integral of the electric field around any closed path, equal to the rate of change of the magnetic flux through any surface bounded by the path.
- One consequence is the current induced in a conducting loop placed in a time-varying $\vec{B}$ field

FARADAY'S LAW IN DIFFERENTIAL FORM

Faraday's law in integral form, $\oint_L \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \iint_A \vec{B} \cdot d\vec{A}$

From Stoke's Theorem we found that, $\oint_L \vec{E} \cdot d\vec{l} = \oiint_A (\vec{\nabla} \times \vec{E}) \cdot d\vec{A}$

Applying Stoke's Theorem to the LHS. We can write,

$$\oiint_A (\vec{\nabla} \times \vec{E}) \cdot d\vec{A} = -\frac{\partial}{\partial t} \iint_A \vec{B} \cdot d\vec{A} \quad \text{Thus, } \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

This is the differential form of Faraday's law.

Ampere-Maxwell Law: This equation combines the effects of electric currents and changing electric fields on the magnetic field. It is expressed as:

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

where $\vec{\nabla} \times \vec{B}$ represents the curl of the magnetic field $\vec{B}$, μ_0 is the vacuum permeability, $\vec{J}$ is the conduction current density, ϵ_0 is the vacuum permittivity, and $\frac{\partial \vec{E}}{\partial t}$ is the partial derivative of the electric field $\vec{E}$ with respect to time.

- In integral form: $\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} + \mu_0 \epsilon_0 \frac{d\Phi_e}{dt}$
- It describes the tendency of the magnetic field to circulate around a region in space with electric current and/or time-varying electric fields.
- The line integral of the magnetic field around any closed path is the given sum.

NOTE: These four equations, together with the Lorentz force law, form a complete set of equations that govern classical electromagnetism. They describe how electric and magnetic fields interact with charges and currents and provide the foundation for understanding a wide range of electromagnetic phenomena, including the behavior of electromagnetic waves, the operation of electrical circuits, and the principles behind many electromagnetic devices.

TRAVELLING ELECTROMAGNETIC WAVE

A traveling electromagnetic wave, also known as an electromagnetic radiation, refers to the propagation of electromagnetic energy through space. Electromagnetic waves consist of oscillating electric and magnetic fields, and they can travel through a vacuum or a medium. These waves encompass a wide range of frequencies and wavelengths, including radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays, and gamma rays.

- Electromagnetic waves are created by the acceleration of electric charges or by the oscillation of electric and magnetic fields.
- These waves consist of two perpendicular components: an electric field and a magnetic field.
- The electric field and magnetic field oscillate in phase with each other and are perpendicular to the direction of wave propagation, forming a transverse wave.
- The wave's energy is transferred perpendicular to the direction of the wave's motion.

Electromagnetic waves can travel through a vacuum, like empty space, as well as through various media, including air, water, and solids. They are characterized by wavelength (λ), frequency (f), and speed (c). The speed of electromagnetic waves in a vacuum is denoted by " c ".

- The electric and magnetic fields are always perpendicular to the direction in which the wave is traveling. Thus, the electromagnetic wave is a *transverse wave*.
- The electric field is always perpendicular to the magnetic field.
- The cross product always gives the direction in which the wave travels.

The electric and the magnetic fields always vary sinusoidally, **with the same frequency** and **in phase** with each other. The electric and magnetic fields as sinusoidal functions can be written as,

$$E = E_m \sin(kx - \omega t) \dots \dots \dots (1)$$

$$B = B_m \sin(kx - \omega t) \dots \dots \dots (2)$$

Electromagnetic waves encompass a wide range of frequencies, known as the electromagnetic spectrum. This spectrum includes various types of waves, such as radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays, and gamma rays. Each type of wave has a different wavelength and frequency, and they are used for various purposes in science, technology, and everyday life, ranging from wireless communication to medical imaging.

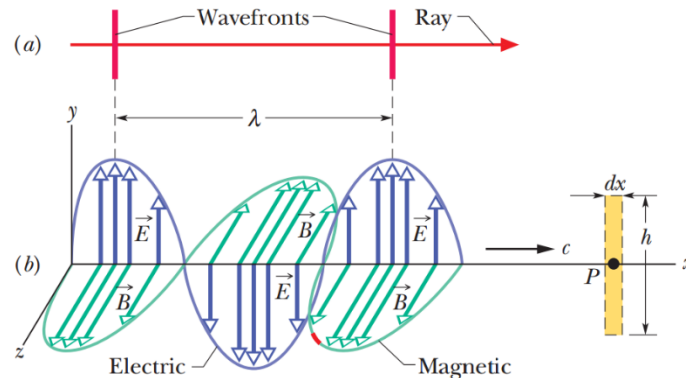


Fig. 1 (a) An electromagnetic wave represented with a ray and two wavefronts; the wavefronts are separated by one wavelength λ . (b) The same wave represented in a "snapshot" of its electric field $\vec{E}$ and magnetic field $\vec{B}$ at points on the x axis, along which the wave travels at speed c . As it travels past point P , the fields vary. The electric component of the wave consists of only the electric fields; the magnetic component consists of only the magnetic fields.

Induced Electric Field:

The dashed rectangle of dimensions dx and h is fixed at point P on the x axis and in the xy plane, is presented in Fig. 1(b).

- As the electromagnetic wave moves rightward past the rectangle, the magnetic flux Φ_B through the rectangle changes and according to Faraday's law of induction induced electric fields appear throughout the region of the rectangle.
- Let us consider, $\vec{E}$ and $\vec{E} + d\vec{E}$ be the induced electric fields along the two long sides of the rectangle. These induced electric fields are, in fact, the electrical component of the electromagnetic wave (Fig. 2).

Let's consider the induced electric fields at the instant when this red portion of the magnetic component is passing through the rectangle.

- The magnetic field through the rectangle points in the positive z direction and is decreasing in magnitude (the magnitude was greater just before the red section arrived).
- As the magnetic field is decreasing, the magnetic flux Φ_B through the rectangle is also decreasing.
- According to Faraday's law, this change in magnetic flux is opposed by induced electric fields, which produce a magnetic field $\vec{B}$ in the positive z direction.

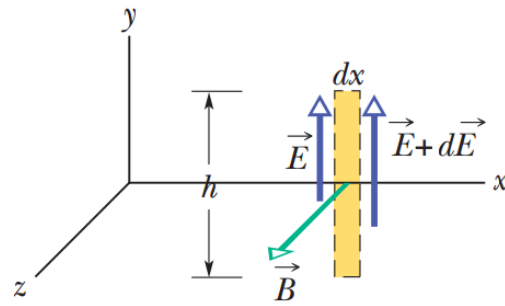


Fig 2. As the electromagnetic wave travels rightward past point P in Fig., the sinusoidal variation of the magnetic field through a rectangle centered at P induces electric fields along the rectangle. At the instant shown, is decreasing in magnitude and the induced electric field is therefore greater in magnitude on the right side of the rectangle than on the left.

- According to Lenz's law, a counterclockwise induced electric field vectors $\vec{E}$ and $\vec{E} + d\vec{E}$ are formed along the two arms of the rectangle, as shown in Fig. 2. The magnitude of $\vec{E} + d\vec{E}$ is greater than that of $\vec{E}$, and, thus, the net induced electric field acts counterclockwise around the rectangle.

Implementing Faraday's Law: From the Faraday's law of induction, we know,

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt} \dots \dots \dots (3)$$

Direction of the induced electric field is counterclockwise around the rectangle (Fig. 2). There is no contribution to the integral from the top or bottom of the rectangle because $\vec{E}$ and $d\vec{s}$ are perpendicular to each other there. The integral then has the value,

$$\oint \vec{E} \cdot d\vec{s} = (E + dE)h - Eh = h dE \dots \dots \dots (4)$$

Here, E is the electric field at one side of the loop, $(E + dE)$ is the electric field at the opposite side of the loop.

Considering a small change dB through the loop, with h and dx being the height and width of the loop, the magnetic flux Φ_B through this rectangle is given by,

$$\Phi_B = BA = (B)(h dx) \quad [\because A = h dx] \dots \dots \dots (5)$$

where B is the average magnitude of $\vec{B}$ within the rectangle and $h dx$ is the area of the rectangle. Differentiating Eq. (5) with respect to t gives

$$\frac{d\Phi_B}{dt} = h dx \frac{dB}{dt} \dots \dots \dots (6)$$

If we substitute Eqs. (4) and (6) into Eq. (3), we find

$$h dE = -h dx \frac{dB}{dt} \dots \dots \dots (7)$$

$$\frac{dE}{dx} = -\frac{dB}{dt} \dots \dots \dots (8)$$

Both B and E are functions of *two* variables, namely, coordinate x and time t . It is assumed that the time (t) is constant while evaluating $\frac{dE}{dx}$, as the Fig. 2 is an “instantaneous snapshot”. Also, while evaluating $\frac{dB}{dt}$, assumption is that the x is constant, because focus is on the time rate of change of B at a particular place, the point P shown in Fig. b. The derivatives under these circumstances are *partial derivatives*, and therefore we can write,

$$\frac{\partial E}{\partial x} = -\frac{\partial B}{\partial t} \dots \dots \dots (9)$$

The minus sign in this equation is appropriate and necessary because, although magnitude E is increasing with x at the site of the rectangle in Fig. 2, magnitude B is decreasing with t .

We have,

$$\frac{\partial E}{\partial x} = kE_m \cos(kx - \omega t) \dots \dots \dots (10)$$

And

$$\frac{\partial B}{\partial t} = -\omega B_m \cos(kx - \omega t) \dots \dots \dots (11)$$

Therefore,

$$kE_m \cos(kx - \omega t) = \omega B_m \cos(kx - \omega t) \dots \dots \dots (12)$$

The ratio ω/k for a traveling wave is its speed, which we are calling c . Equation then becomes

$$\frac{E_m}{B_m} = \frac{\omega}{k} = c \dots \dots \dots (13) \quad \left[\because \omega = 2\pi f ; \quad k = \frac{2\pi}{\lambda} \quad \& \quad c = f\lambda \right]$$

Induced Magnetic Field:

Figure 3 shows another dashed rectangle at point P in the xz plane. As the electromagnetic wave moves rightward past this new rectangle, the electric flux Φ_E through the rectangle changes and according to Maxwell's law of induction induced magnetic fields appear throughout the region of the rectangle. These induced magnetic fields are the magnetic component of the electromagnetic wave.

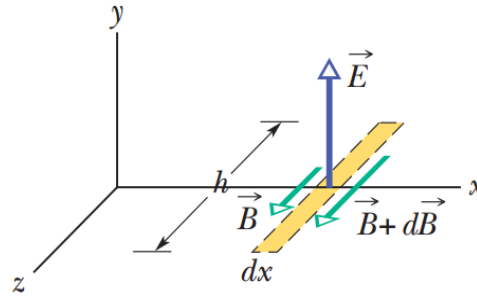


Fig 3. The sinusoidal variation of the electric field through this rectangle, located at point P , induces magnetic fields along the rectangle. The electric field through the loop is decreasing in magnitude, and the magnitude of the induced magnetic field is greater on the right side of the rectangle than on the left.

Let us consider the electric field through the rectangle (Fig. 3) is directed towards position y axis, as shown in figure. This electric field is corresponded with the magnetic field marked in red on the magnetic component curve, where at the chosen instant, the magnetic field is decreasing. As the two fields are in phase, the electric field in Fig. 3 must also be decreasing, and so is the electric flux Φ_E through the rectangle. The changing flux Φ_E induces a magnetic field with vectors $\vec{B}$ and $\vec{B} + d\vec{B}$ oriented as shown in Fig. 3, where field $\vec{B} + d\vec{B}$ is greater than field $\vec{B}$.

Implementing Maxwell's Law: From the Maxwell's law of induction, we get,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \dots \dots \dots (14)$$

by proceeding counterclockwise around the dashed rectangle of Fig. 3. Only the long sides of the rectangle contribute to the integral because the dot product along the short sides of the loop is zero. Thus, we can write,

$$\oint \vec{B} \cdot d\vec{s} = -[(B + dB)h - Bh] = -(B + dB)h + Bh = -h dB \dots \dots \dots (15)$$

Here, B is the magnetic field at one side of the loop, $(B + dB)$ is the magnetic field at the opposite side of the loop.

Considering a small change dE through the loop, with h and dx being the height and width of the loop, the electric flux Φ_E through the rectangle is given by,

$$\Phi_E = EA = (E)(h dx) \dots \dots \dots (16)$$

where E is the average magnitude of $\vec{E}$ within the rectangle. Differentiating Eq. (16) with respect to t gives

$$\frac{d\Phi_E}{dt} = h dx \frac{dE}{dt} \dots \dots \dots (17)$$

Therefore,

$$-h dB = \mu_0 \epsilon_0 \left(h dx \frac{dE}{dt} \right) \dots \dots \dots (18)$$

or, changing to partial-derivative notation as we did for equation,

$$-\frac{\partial B}{\partial x} = \mu_0 \epsilon_0 \frac{\partial E}{\partial t} \dots \dots \dots (19)$$

Again, the minus sign in this equation (19) is necessary because, although B is increasing with x at point P in the rectangle in Fig. 3, E is decreasing with t .

Now,

$$-kB_m \cos(kx - \omega t) = -\mu_0 \epsilon_0 \omega E_m \cos(kx - \omega t) \dots \dots \dots (20)$$

which we can write as

$$\frac{E_m}{B_m} = \frac{1}{\mu_0 \epsilon_0 (\omega/k)} \Rightarrow c^2 = \frac{1}{\mu_0 \epsilon_0} \dots \dots \dots (21) \quad \left[\because \frac{E_m}{B_m} = \frac{\omega}{k} = c \right]$$

Therefore, combining equation we get,

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \dots \dots \dots (22)$$

This the wave speed. Here, $\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A} \approx 1.26 \times 10^{-6} \text{ T.m/A}$, and $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$. Therefore, $c = 299863380.50 \text{ m/s}$.

Electromagnetic Wave Propagation:

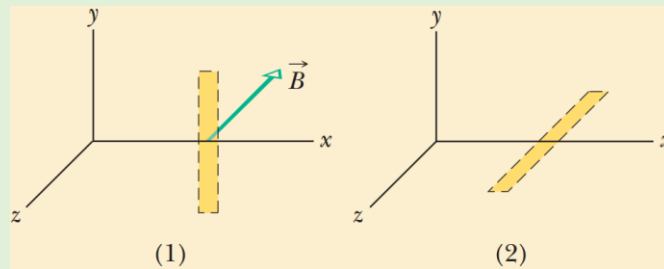
In the context of electromagnetic waves (like light), this relation gives the direction of wave propagation:

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

Here, $\vec{E}$ is the electric field, $\vec{B}$ is the magnetic field, $\vec{E} \perp \vec{B}$ and both are perpendicular to the direction of wave travel, The Poynting vector ($\vec{S}$), is named after John Henry Poynting (1852–1914), an English physicist, represents the direction and rate of energy flow of the wave, and the direction of the pointing vector is given by $(\vec{E} \times \vec{B})$.

- Direction of electromagnetic wave propagation $\vec{c}$ is given by the direction of pointing vector $\vec{S}$.

CHECKPOINT: The magnetic field $\vec{B}$ through the rectangle is shown is directed in the xz plane, parallel to the z axis, and its magnitude is increasing. (a) Complete part 1 by drawing the induced electric fields, indicating both directions and relative magnitudes. (b) For the same instant, complete part 2 of the figure by drawing the electric field of the electromagnetic wave. Also draw the induced magnetic fields, indicating both directions and relative magnitudes.



ANSWER:

(a) On right side of rectangle, $\vec{E}$ is in negative y direction; on left side, $\vec{E} + d\vec{E}$ is greater and in negative y direction (since, induce electric field form in counter-clock wise direction).

(b) $\vec{E}$ is downward increasing. On right side, $\vec{B}$ is in negative z direction; on left side, $\vec{B} + d\vec{B}$ is greater and negative z direction (since, induce magnetic field form in clock wise direction).

